

EV Infrastructure Phase II: Causal and Operational Impacts on Gasoline Use and Air Quality in California

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Abstract

As electric vehicles (EVs) scale in California, understanding their effects on both gasoline demand and air quality is increasingly important for fuel and air-quality policy. Policy certainty and infrastructure growth sharpen this question: EPA approved California's Advanced Clean Cars II waiver on December 18, 2024, locking in a zero-emission sales trajectory (U.S. Environmental Protection Agency, 2024), and the state surpassed 201,180 public/shared charging ports by September 24, 2025 (California Energy Commission, 2025). Evidence also shows a bidirectional relationship between fuel prices and EV adoption: EV sales are 4-6 times more sensitive to gasoline prices than to electricity prices, meaning high gasoline prices spur EV uptake and, conversely, rising EV adoption will displace gasoline consumption. Yet availability alone is not enough: a Bay Area audit found only ~73% of tested DC fast-charging connectors functional, highlighting reliability as a first-order constraint (Rempel et al., 2024).

This study fills these gaps by testing whether the deployment of reliable chargers causally improves local air quality (PM_{2.5}/ozone) and reduces county-level gasoline use as a mediating mechanism. I will exploit two policy levers that transform nominal access into effective access: the federal NEVI standard (effective Feb 28, 2023) requiring >97% average annual uptime per port (Federal Highway Administration, 2023; eCFR, 23 CFR §680.116), and California's AB 2061 reliability reporting regime. For purposes of this analysis, "NEVI-eligible/NEVI-funded" chargers are public DC fast charging stations that meet the NEVI Formula Program's corridor and technical criteria (e.g., located along federally designated Alternative Fuel Corridors and qualifying for NEVI funding under California's deployment plan), rather than all DC fast chargers built post-2023 irrespective of funding source. The NEVI program provides state funding to install public EV fast chargers along designated corridors and requires compliance with location and operational standards; only chargers meeting those criteria and incorporated into state NEVI plans will be coded as NEVI treatment in the causal design. Using county-year panels linking CEC A-15 gasoline sales, NREL AFDC charger records, and EPA AQS air-quality data, I will estimate staggered difference-in-differences/event-study models to quantify effects and equity heterogeneity across income tiers and urban/rural counties.

Purpose of the Study

This project builds on my last summer’s county-level research by moving beyond simple correlations to a causal approach, and by treating operations, such as reliability and real access, as core drivers instead of background details. I’m asking whether program-backed charger rollouts and clear reliability standards actually led to measurable county-level cuts in gasoline use and cleaner air. My primary objective for the research this time is to estimate how infrastructure expansion and operational reliability causally affect per-capita gasoline use and local air quality in California counties, and how those effects differ by income level and urban vs. rural context.

My research questions:

- Do EV charging infrastructure improvements that increase effective access (deployment + reliability) reduce county-year PM2.5 and ozone in California?
- Do those same infrastructure improvements reduce per-capita gasoline sales in the treated counties?
- Does exposure to formal reliability standards — specifically chargers that are subject to the federal NEVI Formula Program’s uptime and reporting requirements and California’s AB 2061 uptime reporting regime — increase the magnitude of air-quality improvements relative to similar charger growth without such policy-linked reliability exposure? For this analysis, *reliability exposure* refers to DC fast chargers that are included in California’s NEVI deployment plans and therefore subject to the NEVI uptime requirements and reporting criteria, as well as those covered under AB 2061 reporting protocols, rather than all chargers built after 2023 irrespective of funding source or operational obligations.
- Do the air-quality and gasoline effects differ by county income tier and urban vs. rural status?

Rationale

California remains a strong proving ground to test whether program-backed charger rollouts and explicit reliability standards reduce gasoline use and improve air quality—and for whom. Policy certainty is high: CARB’s Advanced Clean Cars II requires a full transition to zero-emission new light-duty sales by 2035, and the U.S. EPA approved California’s ACC II waiver on December 18, 2024, which stabilizes long-run demand for charging and provides a consistent policy counterfactual for research (California Air Resources Board, 2022; U.S. Environmental Protection Agency [EPA], 2024). At the same time, California is scaling infrastructure rapidly: by September 24, 2025 the state reported 201,180 fully public and shared charging ports—expanding cross-county and time variation needed to identify causal effects of new supply and higher uptime on fuel displacement and pollutant trends (California Energy Commission [CEC], 2025).

Crucially for a shift from correlation to causation, reliability is now a first-order, measurable operational variable. The National Electric Vehicle Infrastructure (NEVI) final rule requires each charging port to achieve >97% average annual uptime (Federal Highway Administration, 2023), and California’s AB 2061 directs the CEC to define “uptime,” standardize record-keeping, and require semiannual and hourly reliability reporting for covered chargers (CEC, 2024). Together, these rules translate nominal “access” into effective access (sufficient ports, high uptime, spatial reach) and generate policy timing suitable for county-level difference-in-differences or event-study designs. Finally, California’s mix of dense urban corridors and vast rural regions—along with documented disparities in charger access by income and race—makes it well-suited for distributional analysis.

Evidence from California shows lower public-charger access in below-median-income and Black/Hispanic-majority areas (Hsu & Fingerman, 2021), and national analyses show lower accessibility for lower-income groups with substantial geographic heterogeneity (Lou et al., 2024). Field audits in the Bay Area also underscore persistent fast-charging reliability problems—only 73.3% of 655 tested DC fast-charging ports were functional at the time of inspection—reinforcing the need to treat reliability as a core driver, not background noise (Rempel, Cullen, Matteson Bryan, & Vianna Cezar, 2024). By combining policy-driven reliability metrics with spatial sufficiency (e.g., ratio and reach) and measuring co-pollutants with EPA’s Air Quality System (AQS), this study can test whether reliable, equitable expansion leads to within-county declines in gasoline use alongside contemporaneous improvements in PM_{2.5}/ozone (EPA, 2025).

Improving air quality is a central motivation for California’s EV policies, given the state’s historic struggles with smog and particulate pollution. Notably, emerging empirical evidence confirms that EV adoption can yield tangible air-quality and health benefits. A recent nationwide study (Baran et al., 2025) finds that greater EV uptake significantly reduces tailpipe pollutants and improves health outcomes – for example, a one-standard-deviation increase in EV registrations led to roughly a 4% decline in NO₂ concentrations, a 2.6% drop in very low birth weight incidents, and an 11% reduction in child asthma emergency visits. These findings underscore the real-world environmental payoffs of vehicle electrification, reinforcing the importance of our inquiry in California. Our study will build on and complement this evidence by examining the specific role of infrastructure reliability standards and their effect on local PM_{2.5} and ozone levels. Furthermore, California’s mix of dense urban corridors and vast rural regions – along with documented disparities in charger access by income and race – makes it well-suited for a distributional analysis of these outcomes. Prior evidence shows lower public-charger availability in below-median-income and majority Black/Hispanic areas of California (Hsu & Fingerman, 2021), and national analyses find lower EV charging accessibility for lower-income groups with substantial geographic heterogeneity (Lou et al., 2024). By focusing on whether reliability improvements can narrow such gaps, we directly address equity concerns in the transition to cleaner transportation. In sum, California offers both the policy variation and the equity imperative to rigorously test whether enhanced EV infrastructure yields cleaner air and who benefits from these gains.

Literature review

The relationship between electric-vehicle (EV) adoption, charging infrastructure, and gasoline consumption has progressed from descriptive correlations to questions of causal impact, especially as California scales chargers and formalizes reliability standards. In December 2024, the U.S. Environmental Protection Agency (EPA) approved California’s Advanced Clean Cars II waiver, confirming the state’s zero-emission sales trajectory and providing a stable policy backdrop for empirical designs (U.S. Environmental Protection Agency [EPA], 2024). By September 24, 2025, California reported 201,180 public and shared charging ports—substantially expanding the cross-county and time variation needed for quasi-experimental work (California Energy Commission [CEC], 2025).

Impact of EV Adoption on Gasoline Consumption

Some empirical literature links higher EV adoption to lower gasoline demand. Using California EV registrations, Bushnell, Muehlegger, and Rapson (2022) show that EV sales are four-to-six times more sensitive to gasoline prices than to electricity prices, exploiting a border-discontinuity in residential electricity tariffs and panel fixed effects—evidence that policy and price shocks that accelerate EV uptake plausibly displace gasoline use (Bushnell et al., 2022). In a county-year fixed-effects analysis of California (2014–2023), *Van* (2025) finds that increased Level 2 chargers and a higher

charger-to-EV ratio are associated with lower per-capita gasoline use, while raw DC fast charger counts show a counterintuitive positive correlation—underscoring how nominal counts can mask congestion, siting, or equity frictions. The model exhibits high cross-sectional explanatory power (adjusted $R^2 \approx 0.89$) but low within-county R^2 , suggesting infrastructure alone explains only a modest share of year-to-year changes, and motivating more operations-focused measures of “effective access.”

Infrastructure, Reliability, and “Effective Access”

Counts alone overstate usable access: reliability (uptime) and spatial reach condition whether chargers actually displace gasoline. A Bay Area field audit found only 73.3% of 655 DC fast charger connectors were functional at the time of testing, with failures concentrated in payment, screen, network, and connector issues (Rempel et al., 2023). In response, policy now treats reliability as a first-order requirement: the federal NEVI rule mandates >97% average annual uptime per port (Federal Highway Administration, 2023), and California’s AB 2061 directs the CEC to define “uptime,” standardize recordkeeping, and require periodic reliability reporting—creating measurable “effective access” and policy shocks suitable for DiD and event-study designs (CEC, n.d.).

Environmental Co-Benefits: PM2.5 and Ozone

Modeling and empirical studies generally find that electrification produces improvements in PM2.5 and ozone in California, with reductions scaling with EV fleet turnover (Skipper et al., 2023). Empirically, recent work has begun to observe these benefits in real-world data. Baran et al. (2025) provide the first nationwide causal evidence that EV adoption measurably improves air quality and health outcomes. Recent work in *Nature Communications* documents that growing ZEV adoption narrows PM2.5 exposure gaps between disadvantaged and non-disadvantaged communities, highlighting measurable local air-quality gains and equity-relevant trends (Yu et al., 2023). These results justify pairing gasoline outcomes with EPA AQS PM2.5/ozone measures in a panel design to test for contemporaneous co-benefits (EPA, 2025a, 2025b).

Equity, Access, and “Charging Deserts”

Multiple studies document disparities in charger access across income and race, especially where multi-unit dwellings are prevalent. In California, Hsu and Fingerman (2021) show lower public-charger access in below-median-income and Black/Hispanic-majority block groups. Nationally, Lou et al. (2024) find lower-income groups are, on average, farther from public EV infrastructure—patterns with significant spatial heterogeneity across counties. These findings motivate heterogeneity tests by income tier and urban/rural status in California counties.

Methods and Data Landscapes for Causal Identification

California’s policy timing (ACC II waiver), scale-up in ports (201,180 as of Sept. 24, 2025), and enforceable reliability metrics (NEVI 97% uptime; AB 2061 reporting) create staggered variation well-suited to difference-in-differences and event-study estimators that distinguish reliable deployment from nominal capacity. The CEC’s infrastructure statistics and reliability proceedings, combined with EPA AQS monitoring, furnish the data backbone for county-level panel designs with contemporaneous air-quality outcomes (CEC, 2025; CEC, n.d.; Federal Highway Administration, 2023; EPA, 2025b).

Conclusion and Research Gap

In sum, the literature supports three pillars my study will connect causally: EV uptake depresses

gasoline demand (Bushnell et al., 2022); reliability and spatial reach condition effective access and usage (Rempel et al., 2023; Federal Highway Administration, 2023; CEC, n.d.); and local PM2.5/ozone improvements are plausible and measurable as electrification proceeds (Skipper et al., 2023; Yu et al., 2023). What's missing is solid county-level evidence that reliable, program-driven charger deployment actually lowers gasoline consumption and improves PM2.5 and ozone—and how those effects vary by equity. This study targets that gap, leveraging California's latest policy and reliability changes (EPA, 2024; CEC, 2025; Federal Highway Administration, 2023).

Research Design

Overview

I will estimate the causal effect of EV charging deployment—emphasizing effective access (counts + reliability policy exposure)—on per-capita gasoline use and local air quality. The design exploits the staggered rollout of reliable fast chargers and the introduction of the federal NEVI uptime standard in early 2023. Not all counties received NEVI-funded corridor chargers at the same time, which creates variation in treatment timing that is crucial for identification. I will use modern staggered difference-in-differences and event-study estimators that are robust to heterogeneous treatment timing (Callaway & Sant’Anna, 2021; Sun & Abraham, 2021). The NEVI reliability rule (which requires >97% uptime from 2023 onward) effectively converts nominal charger access into effective access, providing a plausibly exogenous shock at the county level. This policy timing, with some counties seeing reliable charger deployment in 2023 and others in 2024, enables within-county comparisons before vs. after reliable infrastructure arrives, as well as between-county comparisons of treated vs. not-yet-treated areas. By focusing on NEVI exposure as the treatment, we ensure the analysis can be executed with data through 2024 (yielding two post-treatment years) while still capturing the reliability-induced “NEVI effect.” The identification framework remains a difference-in-differences design, supplemented by event-study diagnostics to check pre-trends and to observe how outcomes evolve after treatment.

In this study, a “NEVI charger” is defined as a public DC fast charging station that meets the criteria for participation in the federal National Electric Vehicle Infrastructure (NEVI) Formula Program and California’s associated deployment plan. Under NEVI guidance, eligible chargers are typically situated along designated Alternative Fuel Corridors (AFCs) and meet technical and operational standards (e.g. DC fast charging capability, network connectivity, and reporting). NEVI funding and eligibility does not automatically come simply because a charger is built after 2023. Only chargers that satisfy the location and technical requirements in California’s NEVI deployment plan and the Federal Highway Association (FHWA)’s NEVI guidance are *counted as NEVI-eligible in this analysis*. Chargers outside these criteria (e.g., stations not on AFCs or not built under the NEVI deployment process) remain treated as non-NEVI infrastructure in the causal design.

Data

County gasoline sales (annual): California Energy Commission CEC-A15 workbook with county tables for gasoline sales & station counts, 2010–2024. (Note: A-15 covers ~86% of statewide gasoline sales.)

Charging stations (locations & attributes): AFDC Alternative Fuel Stations API (fields incl. coordinates, open_date, status, EVSE/DCFC counts). Used to build ports per 1,000 residents, DCFC share, and first-year DCFC presence by county.

NEVI corridor overlay (eligibility flag): California Alternative Fuel Corridors / NEVI Corridor Group GIS layers to tag new DCFC openings as on-corridor vs. off-corridor.

Air-quality outcomes (daily → county-year): EPA AQS daily PM_{2.5} and ozone downloads, aggregated to county-year to align with gasoline outcomes.

Population denominator (per-capita rates): U.S. Census ACS county Total Population (Table B01003), latest 5-year estimates.

Income stratifier (equity splits): U.S. Census ACS county Median Household Income (Table B19013), 5-year estimates.

Urban/rural stratifier: USDA ERS Rural-Urban Continuum Codes (RUCC), county-level.

Variables

Variable	Description	Unit	Source
Gasoline use per capita	County gasoline sales normalized by population	Gallons per person (annual)	CEC A-15; US Census ACS population
Annual county PM2.5	Aggregate daily PM2.5 to county-year	$\mu\text{g}/\text{m}^3$ (annual)	EPA AQS
Annual county ozone	Aggregate EPA AQS daily ozone to county-year	Ppm (annual)	EPA AQS
Number of ports per 1000 residents	Charging availability intensity	Ports per 1000 people (annual)	AFDC Stations data, ACS population
DC fast charging share	Proportion of DC fast charging by year	Proportion (0-1)	AFDC Stations data
NEVI corridor DC fast charging additions	Number (or per-capita) of DC fast charging ports that meet federal NEVI Formula Program criteria	Ports	AFDC Stations API, NEVI corridor layers.
Post-NEVI policy time indicator	Time indicator for federal reliability rule	Binary (0 before the policy, 1 after)	Federal Register.
Median household income	ACS county median household income	US dollars	US Census ACS
Urban/rural	USDA Economic Research Service (ERS) Rural-Urban Continuum Code for county	RUCC category	USDA ERS RUCC

Models

Model 1: Effect of reliable charging on gasoline use

I will build this model to examine the impact of charger expansion and reliability policies on gasoline use, which we treat as a key intermediate outcome linking EV adoption to emissions. The outcome variable is annual *per-capita gasoline consumption* in county c and year t .

To address the concern that there are only two post-treatment years (2023–2024), I implement NEVI exposure as a **continuous (dose) treatment** in a DiD framework, rather than a single binary on/off indicator. Intuitively, even within a short post-period, counties can experience **different magnitudes of NEVI-relevant corridor fast-charger expansion**, and this “dose” variation provides additional identifying variation for the DiD estimate.

Formally, I will estimate specifications of the form:

$$GasUse_{ct} = \alpha + \beta_1 (NEVIDose_{ct} \times PostNEVI_t) + \beta_2 PortsPerCapita_{ct} + \beta_3 DCFastShare_{ct} + \gamma_c + \delta_t + \epsilon_{ct}$$

where $GasUse_{ct}$ is gallons of gasoline per person in county c and year t . Here, $PostNEVI_t$ equals 1 in years 2023 and later (after the NEVI reliability rule took effect) and 0 before 2023. The key DiD regressor is the interaction $NEVIDose_{ct} \times PostNEVI_t$, where $NEVIDose_{ct}$ is a county-year measure of NEVI exposure intensity (the “dose”), constructed from observed NEVI-corridor DC fast charging buildout. Concretely, this can be measured as NEVI-corridor DCFC ports per capita (e.g., ports per 1,000 residents) or equivalently as the annual change in NEVI-corridor DCFC ports per capita. The key requirement is that this variable varies across counties and over time and reflects the magnitude of NEVI-relevant corridor fast-charger expansion. The interaction $NEVIDose_{ct} \times PostNEVI_t$ therefore captures a difference-in-differences-with-intensity treatment: it asks whether gasoline use falls more (after 2023) in counties that experience larger increases in NEVI corridor fast-charging availability, relative to counties with little or no NEVI-relevant buildout over the same period. Under this interpretation, β_1 is the average within-county change in per-capita gasoline consumption after 2023 for counties with corridor chargers (relative to those without). We include $PortsPerCapita_{ct}$ and $DCFastShare_{ct}$ as time-varying controls for the general availability and composition of charging infrastructure, since even in the absence of the policies, growing charger density might reduce gasoline use. County fixed effects γ_c absorb all time-invariant differences between counties (e.g. geographic and socioeconomic factors affecting fuel use), and year fixed effects δ_t absorb statewide shocks (e.g. macroeconomic trends or statewide fuel price changes affecting all counties). Standard errors will be clustered at the county level to account for serial correlation within counties.

Although this baseline two-way fixed effects specification provides an initial estimate, we will further leverage the staggered adoption nature of NEVI exposure to get more nuanced estimates. Because counties can begin experiencing non-zero NEVI corridor DCFC “dose” at different times (e.g., first

meaningful increase in 2023 vs. 2024), we can define treatment timing based on the first year a county crosses a meaningful NEVI_Dose threshold (or the first year NEVI_Dose becomes non-zero) and then apply modern DiD methods. Specifically, we will use the Callaway & Sant'Anna (2021) framework to estimate group-time average treatment effects for cohorts defined by first exposure timing, and the Sun & Abraham (2021) approach to implement an event-study analysis that traces out dynamic effects before and after a county's first meaningful NEVI exposure. The event-study will allow us to check the parallel trends assumption and to observe how quickly and to what extent gasoline consumption changes after counties begin receiving NEVI-relevant corridor fast-charger expansion. In sum, Model 1 identifies the causal impact of NEVI-standard charger expansion intensity on gasoline usage, while also exploring timing and persistence.

Model 2: DiDs for air-quality co-benefits

I will use DiDs to test whether gasoline reductions coincide with local PM2.5/ozone improvements. The dependent variables will be the annual county-level PM2.5 and ozone constructed from EPA AQS daily data aggregated to county x year. I will use the same independent variables in this model as in the model 1.

Research Work Plan

Week	Objectives	Activities
1	Literature review and finalize approach	Read and synthesize the key papers on EV adoption, reliability and equity. Confirm the policy anchors (NEVI uptime). Finalize hypotheses, and treatment definitions and write a short pre-analysis plan.
2	Data collection	Download CEC A-15, AFDC stations, NEVI corridor GIS layers, EPA AQS daily PM2.5 and ozone, ACS population and income, and USDA ERS rural-urban code.
3	Build analysis variables	Create per-capita gasoline and county-year PM2.5/ozone. Construct other variables based on collected data. Produce a tidy county-year panel and a data dictionary.
4	Explore and sanity check	Check for missing values, clean the data. Make plots to spot any trends for the DiD/event-study work.
5	Model 1	Run the DiD and event study with county/year fixed effects. Cluster errors at county level. Review pre-trends and find where effects show up.
6	Model 1	Run robustness checks: compare a ‘first-DCFC-year’ spec (FirstDCFC) to an ‘intensity’ spec, try alternative ways to define what counts as an on-corridor site, re-weight results by county population.
7	Model 2	Aggregate AQS to county-year and mirror Model A for PM2.5 and ozone. Compare timing and magnitudes to the gasoline results. Export clean tables and event-study figures.
8	Equity lens	Add interactions for income tiers and urban-rural code. Summarize distributional effects in a couple of clear tables and simple county maps, tied back to literature.
9	Write-up	Putting all sections together, draft the paper.
10	Finalize paper, poster	Finalize the research paper, come up with an outline for the poster.

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